

PAPER CODE	EXAMINER	DEPARTMENT	TEL
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2nd SEMESTER 2024/25 Individual Report Brief

BSc Information Management and
Information Systems

Business Intelligence

INSTRUCTIONS TO CANDIDATES

1. A single copy of this report must be submitted electronically through Learning Mall Online by **11:59 pm, Sunday, 25 May 2025** (China Standard Time, UTC/ GMT+8).
2. The report should be less than 3,000 words. This does not include appendices (if needed), but the appendices should not be longer than 10 pages.
3. This report will count for 70% of the final mark in this course.

Coursework: Individual Report**Context**

Credit score cards are a common risk control method in the financial industry. It uses personal information and data submitted by credit card applicants to predict the probability of future defaults and credit card borrowings. The bank can decide whether to issue a credit card to the applicant. Credit scores can objectively quantify the magnitude of risk.

Credit score cards are based on historical data. Once encountering large economic fluctuations, past models may lose their original predictive power. Logistic model is a common method for credit scoring, because Logistic is suitable for binary classification tasks and can calculate the coefficients of each feature. To facilitate understanding and operation, the score card will multiply the logistic regression coefficient by a certain value (such as 100) and round it.

At present, with the development of machine learning algorithms. More predictive methods such as Boosting, Random Forest, and Support Vector Machines have been introduced into credit card scoring. However, these methods often do not have good transparency. It may be difficult to provide customers and regulators with a reason for rejection or acceptance.

Task

Build a machine learning model to predict if an applicant is a 'good' or 'bad' client, different from other tasks, the definition of 'good' or 'bad' is not given. You should use some techniques to construct your label. Also, unbalance data problem is a big problem in this task.

Data Content & Explanation (source: www.kaggle.com)

There're two tables in this case that could be merged by ID:

application_record.csv

Feature name	Explanation	Remarks
ID	Client number	
CODE_GENDER	Gender	
FLAG_OWN_CAR	Is there a car	
FLAG_OWN_REALTY	Is there a property	
CNT_CHILDREN	Number of children	
AMT_INCOME_TOTAL	Annual income	
NAME_INCOME_TYPE	Income category	
NAME_EDUCATION_TYPE	Education level	
NAME_FAMILY_STATUS	Marital status	

application_record.csv

NAME_HOUSING_TYPE	Way of living	
DAYS_BIRTH	Birthday	Count backwards from current day (0), -1 means yesterday
DAYS_EMPLOYED	Start date of employment	Count backwards from current day(0). If positive, it means the person currently unemployed.
FLAG_MOBIL	Is there a mobile phone	
FLAG_WORK_PHONE	Is there a work phone	
FLAG_PHONE	Is there a phone	
FLAG_EMAIL	Is there an email	
OCCUPATION_TYPE	Occupation	
CNT_FAM_MEMBERS	Family size	

credit_record.csv

Feature name	Explanation	Remarks
ID	Client number	
MONTHS_BALANCE	Record month	The month of the extracted data is the starting point, backwards, 0 is the current month, -1 is the previous month, and so on
STATUS	Status	0: 1-29 days past due 1: 30-59 days past due 2: 60-89 days overdue 3: 90-119 days overdue 4: 120-149 days overdue 5: Overdue or bad debts, write-offs for more than 150 days C: paid off that month X: No loan for the month

Alternative data source:

If you don't want to use this topic and dataset, you may also select an alternative data set from other data sources, for example (but not limit to):

1. <https://www.kaggle.com/>
2. <https://aistudio.baidu.com/competition> (you must present in English)
3. <http://archive.ics.uci.edu/ml/>

Requirements:

Your research report will be a comprehensive analysis of a single topic on business intelligence

and data analytics. It would be the same topic that you have presented in group presentation. You will need to carefully examine why the topic chosen is important, the research that has been done by others, the data source / data set you select, the data exploration and analysis process, the results and conclusion. This report will count for 70% of your final mark. There are 100 total marks available.

- The report must be less than 3000 words. This does not include appendices (if needed), but the appendices should not be longer than 10 pages. This will require you to demonstrate the succinct (compact) writing skills that are required in writing executive summaries in the industry.
- The report should have page numbers inserted.
- The report must be submitted by **11:59 pm, Sunday, 25 May 2025 (China Standard Time, UTC/ GMT+8)** on Learning Mall Online.
- This should be structured as an academic style research paper using **Harvard referencing** (<https://libguides.lib.xjtlu.edu.cn/xjtlureferencing>). There are several examples of such style already provided throughout the course. You may want to consult with the Language Centre for help with this as well.
- Plagiarism will not be tolerated and will be dealt with in the standard way. Your report should include the URLs of the data set and any other on-line sources that you have used.

Report structure:

The report may have different structure. A sample structure is as below:

Title Page

Table of Contents

- 1. Executive Summary**
- 2. Introduction / case background**
- 3. Relevant research / literature review**
- 4. Data understanding / exploration / preparation**
- 5. Modelling / methods / data analysis**
- 6. Results interpretation / discussion**
- 7. Conclusion, limitation and future work**
- 8. References**
- 9. Appendices (if any, such as the core code)**

Some general issues:

- In a case study it is crucial that you integrate relevant theory from the course and evidence from the case. Failure to attempt to integrate theory will lead to severe mark reduction or failure.
- Referencing of all non-original material is essential. You will lose marks for poor referencing.
- Check your completed work for internal consistency.
- Try not to be overly descriptive. Remember you are trying to identify, analyze, and solve the problems of the case using the relevant theories from the course - not just repeating what

the references, or case information, has said.

Marking:

The assessment will comprise of the following criteria:

1. Report presentation (20%)

The extent to which the assignment represents an effective report. This will be judged on:

Appearance: Is a word count included at the end of the report? Is it within the specified amount?

Structure: Does the report follow the conventions of the format? Does it have a clear introduction, explaining how it answers the questions? Do the sections of the report develop ideas in a logical sequence?

Spelling and grammar: Are all words spelled correctly and is the meaning of sentences clear?

Referencing: Have appropriate references been included in the report. Has a recognized referencing system been used for notation? Harvard-style referencing is required here.

2. Use of relevant theory (20%)

Has the right theoretical content / modelling method been chosen as the basis for solving the problem? Is the theory that is selected significant to the questions?

3. Analysis Process (40%)

This measures the extent to which students conduct a systematic data analysis process, by combining relevant modelling method / theory with the information provided in the case. The process includes data understanding, preparation, exploration, modelling, analysis, etc. Other evaluation criteria will be the quality and accuracy of the research and analysis; preparedness, usefulness, creditability.

4. Result interpretation / conclusion (20%)

This measures the presentation of the results in an elegant and very informative way, including the use of visualization methods to show the results clearly, the evaluation of the model, and managerial recommendation.

Grading Criteria:

First Class (70 – 100%)

Makes many insightful connections between real world situations and theoretical considerations.

Makes a thorough study on the selected topic.

Explores and analyses the questions in an in-depth way.

Presents in an elegant and very informative way.

Demonstrates the thought clearly.

Writes fluently, with evidence of a highly developed capacity to structure work systematically and argue logically.

Upper Second Class (60 – 69%)

Makes an appropriate study on the selected topics.

Explores and analyses the topic following a systematic procedure. Comprehensive knowledge of concepts and theories.

Employs appropriate application of theory and experience to the questions.

Demonstrates ability to inter-related concepts and ideas.

Shows some originality in approach and awareness of scope and limitations.

Answers questions in a systematic and coherent manner.

Lower Second Class (50-59%)

Shows evidence of knowledge of concepts and theories.

Explores and analyses the questions, but lacking depth and/or details.

Addresses the main issues appropriately.

Presents the answers clearly without error.

Presents the work in a structured form but arguments are weak in places.

Third Class (40-49%)

Shows Evidence of uncritical knowledge of main concepts and theories.

Attempts to relate theory to practice with limitations and relies on personal opinion or assertions.

Shows limited evidence of data analytical or theoretical skill.

Presentations and structure are weak in several places.

Fail (0 – 39%)

Shows some knowledge of main concepts and theory but major omissions and/or misunderstandings.

Styles and structure are weak and overly descriptive.

Has considerable limitations in ability to present analysis.